TEAM VBALL · SDC HACKATHON 2026 · DOMAIN 2

SMErgers

Agreed on WhatsApp. Billed from memory.
Not any more.

01
Catch it

02
Confirm it

03
Sync it

Orbyte Smart SME Challenge
SDG 8.2 · 8.3

PROBLEM

Agreed on WhatsApp. *Lost at billing.*

Every change nobody writes down is money the company absorbs.

35

WhatsApp messages in one dinner’s group chat

Synthetic dinner job · RM34,530 quote

10

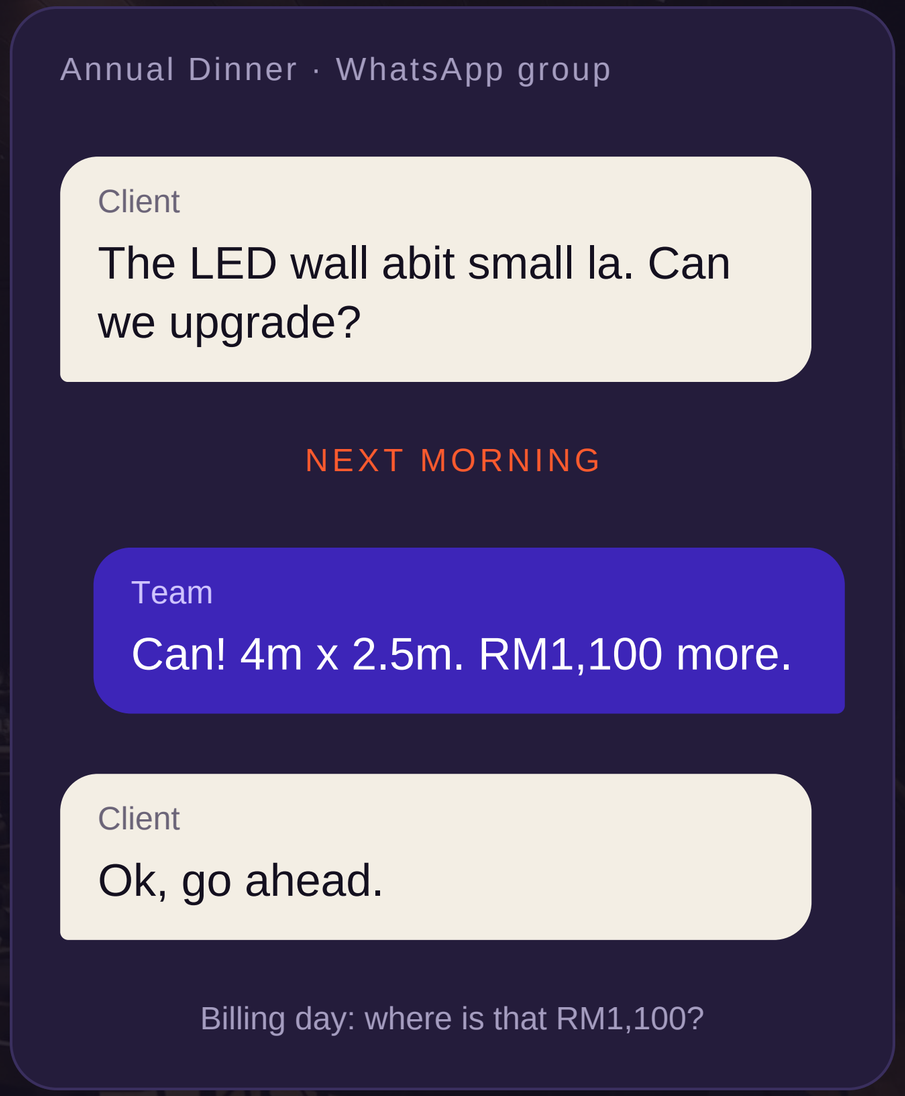
changes hidden in them, none written into the signed quote

SMErgers links each to its messages

11.2%

RM3,860 on that quote, agreed in chat and recorded nowhere

Rebuilt from memory at billing time



A synthetic example job, written from a team member’s experience at an event company, where this happened on almost every event. No public data exists for event companies, so the pilot measures the real rate.

PROBLEM

Where the money disappears

1

Quote signed

RM34,530 for 180 guests

2

Changes in WhatsApp

+15 guests, bigger LED, overtime

3

Bill from memory

Someone scrolls weeks of chat

4

“I never agreed.”

No record to show them

5

Owner gives in

Extras written off to keep the client

Nobody is careless. The agreement just lives in *a chat, not a system.*

PROBLEM

Nobody turns the chat into the bill

CAN IT…	**WhatsApp + memory**	**Accounting apps** Bukku, AutoCount	**Event software** Rentman, Event Temple	**Contractor tools** CoConstruct, Baukraft	SMErgers
Read agreements from WhatsApp	×	×	×	**Partly**	✓
Match each change to the quote	×	×	**Partly**	✓	✓
Client confirms each change	×	×	**Partly**	✓	✓
Post the new version to the group	×	×	×	×	✓
Link each bill line to its chat proof	×	×	×	×	✓

Public pages and search results, 21 Oct 2026. Partly = a limited form. × = not found, not proof it does not exist.

SOLUTION

Catch it. *Confirm it.* Sync it.

CATCH IT



Messages arrive

From the job’s WhatsApp group



Chat analysed

AI reader finds item, quantity, price, source

CONFIRM IT



PM checks

Adds any missing price



Director signs off

RM500 or more, and discounts



Client confirms

From a link, no login

SYNC IT



Quote v2 to the group

Plus bill, CSV, log

Nothing reaches the bill without a person’s check and the client’s yes.

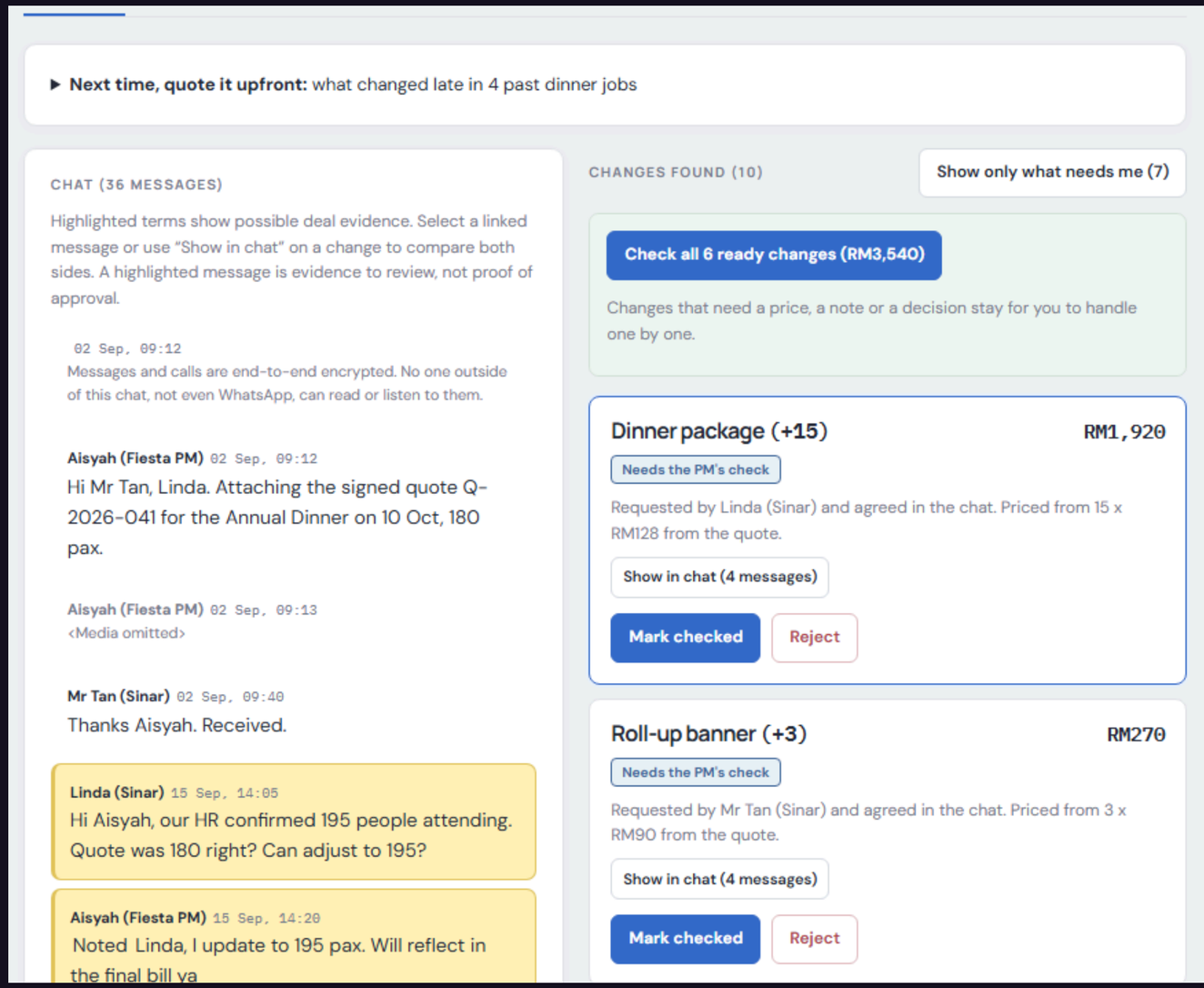
SOLUTION

The project manager's view

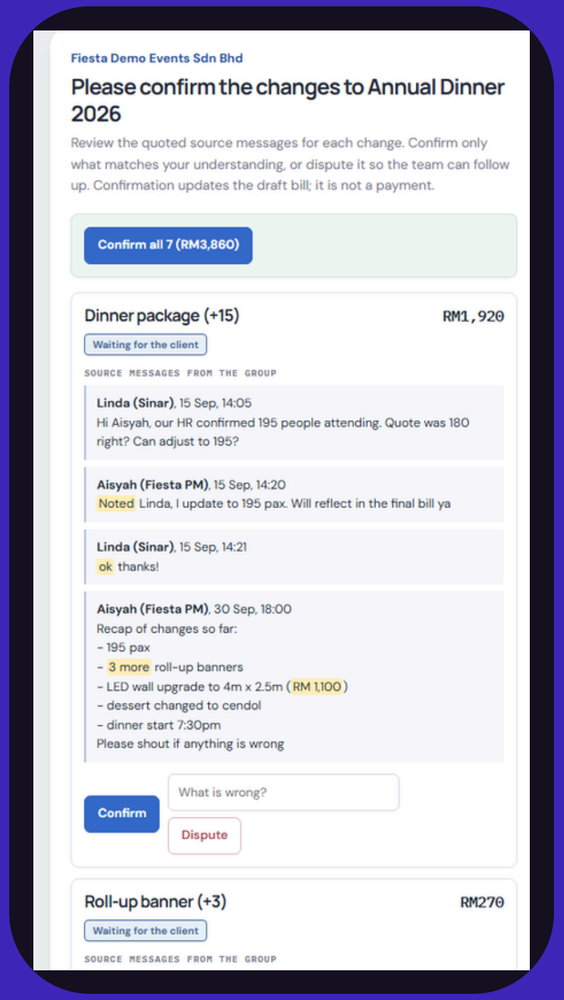
1 **Know what waits for you**
RM3,540 agreed, 7 changes to check

2 **Every change has proof**
The client's own words light up

3 **Check six in one click**
Unpriced items wait for a person



Screenshot of the working prototype, synthetic data.



1 · Client taps the link

SOLUTION

The client confirms. Everyone stays in sync.

Annual Dinner · WhatsApp group

SMErgers

Confirmed by Sinar Demo Trading Sdn Bhd:

- Dinner package (+15): RM1,920
- LED wall 3m x 2m, changed to 4m x 2.5m: RM1,100
- Emcee overtime (+1): RM300
- and 4 more

Quote Q-2026-041 is now version 2, total RM38,390.

Version 1

RM34,530

Version 2

RM38,390

2 · The group is told. Demo: synthetic group. Production: WhatsApp Business API.

Screenshot and message are from the working prototype, synthetic data.

SOLUTION

People stay in control

Who	Can	Cannot
Project manager	Check, add a missing price, reject	Approve big items or discounts
Project director	Approve RM500 or more, decide discounts	Confirm for the client
Client	Confirm, or dispute with a reason	See the team’s pages

HUMAN APPROVAL

Nothing is billed without the PM’s check and the client’s yes.

CHECKED ON THE SERVER

Not just hidden buttons. A client link can’t approve its own change.

ACTIVITY LOG

Who, role, what, when. Each entry is chained, so an edit shows.

Sign-in is simulated, as the brief allows. The RM500 limit is a setting.

TECHNICAL APPROACH

How it works

TRIGGER	DATA	CHAT ANALYSIS	APPROVAL	OUTCOME
WhatsApp Business API Signed webhook	Quote and rate card Directory, past jobs	Reads the chat AI reader, rules as backup	PM, director, client Roles enforced on the server	Quote v2 and bill Group update, CSV, API, log

AI WHERE IT HELPS

AI reads the messy chat. Rules and people decide. On blind chats it beat our rules: 17 of 29 changes right, against 25.

ORBYTE-READY

Reuses its roles, approvals, eSign and audit trail. Adds chat to approved changes.

BUILT AND TESTED

Python and Flask on Replit. 290+ automated tests. All data synthetic.

Labels are the five steps of the Domain 2 brief. Demo runs on synthetic WhatsApp messages.

TECHNICAL APPROACH

Plugs into Orbyte. *Adds what it lacks.*

SMErgers needs	Orbyte provides
The job and its client	Clients and projects
Who is PM, director, client	Directory, shared roles
Director and client sign-off	Approvals, eSign
A record nobody can edit	Audit trail
Sending the bill out	Accounting partners
Job data for other modules	Open APIs

ADDS, DOES NOT COPY

Chat turned into approved changes

REUSE, SHOWN

A renovation chat runs unchanged. Only the quote file differs.

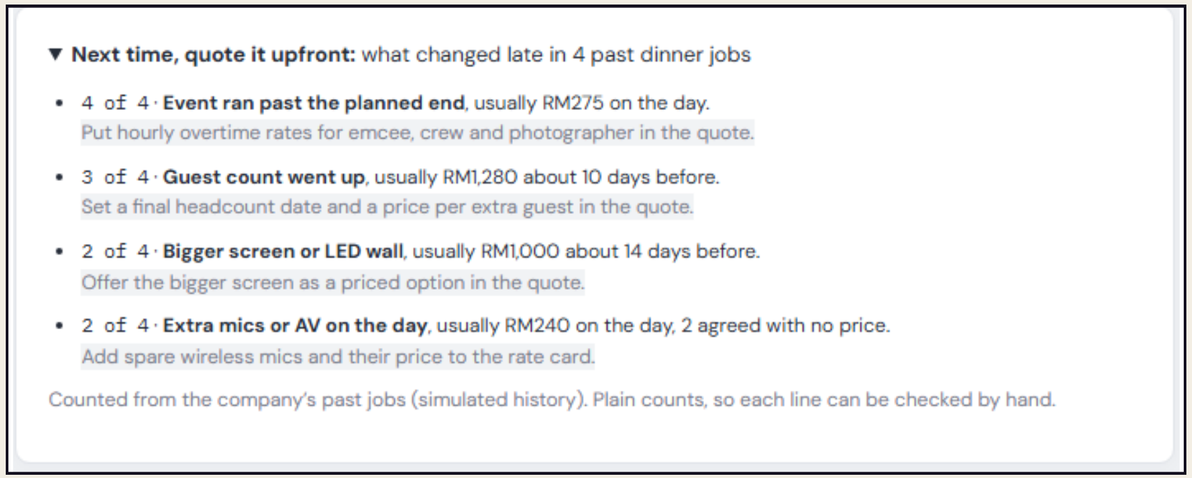
BUILDING BLOCK

A job API returns changes, approvals, bill and log.

Orbyte modules as described in the organizers' Orbyte introduction. Integration points are modelled in docs/orbyte-integration.md.

TECHNICAL APPROACH

It learns from every job



▼ **Next time, quote it upfront:** what changed late in 4 past dinner jobs

- `4 of 4` · **Event ran past the planned end**, usually RM275 on the day.
  Put hourly overtime rates for emcee, crew and photographer in the quote.
- `3 of 4` · **Guest count went up**, usually RM1,280 about 10 days before.
  Set a final headcount date and a price per extra guest in the quote.
- `2 of 4` · **Bigger screen or LED wall**, usually RM1,000 about 14 days before.
  Offer the bigger screen as a priced option in the quote.
- `2 of 4` · **Extra mics or AV on the day**, usually RM240 on the day, 2 agreed with no price.
  Add spare wireless mics and their price to the rate card.

Counted from the company's past jobs (simulated history). Plain counts, so each line can be checked by hand.

SYNTHETIC DEMO HISTORY

4 of 4

past dinners in our demo ran past the planned end.
Put overtime in the quote.

NOW

Plain counts from past jobs. Fewer last-minute decisions.

NEXT

Every approval is a labelled example. After a pilot, it trains a model.

Built on 10 synthetic past jobs of one demo company. A model is trained on real approvals after a pilot.

BUSINESS

Who it is for

Micro and small businesses that run on WhatsApp



START HERE

Event companies

Dinners, launches, weddings. Changes fly in WhatsApp.

NEXT

Renovation, contractors

Same flow: extras agreed on site, then argued.

THEN

Agencies, print, IT

Any team that bills by the job and scopes in chat.

BUYER

The owner. Loses money to unbilled extras.

DAILY USER

The project manager. Chases details across chats.

TOUCHPOINT

The client. Taps a link. No login.

Photo: Unsplash.

BUSINESS

Business model

B2B · freemium subscription (SaaS)

CAPTURE

RM0

always free

One job at a time. Find and list every change.

CONFIRM

RM100

per company, per month

Approvals, client links, quote updates to the group.

SETTLE

RM250

per company, per month

All jobs. Draft bills, CSV, suggestions, log export.

Free to start. It pays for itself with the first extra it recovers.

B2B: businesses pay. Their clients never do.

Channel: Orbyte, then event associations. Option: usage credits.

Launch prices, validated with owners in the pilot.

BUSINESS

Market size



TAM · PROJECT-BASED SMES

≈128,000 firms × RM1,200 a year

Renovation and contracting 102,657 firms (DOSM 2023) + about 25,000 events, agencies, print (assumption)

SAM · RUN JOBS OVER WHATSAPP

≈25,500 firms (20% of TAM, assumption)

3 or more staff, jobs billed by project

SOM · WITHIN THREE YEARS

≈510 firms (2% of SAM, assumption)

Event companies first, through Orbyte and associations

Sources and every assumption: appendix A. No event-company count exists in public data.

BUSINESS

Does it pay for itself?

1 · EXTRAS A YEAR

RM38,600

RM3,860 on our dinner, times 10 dinners a year

→

2 · WON BACK

RM9,650

We win back one quarter of those extras

−

3 · SUBSCRIPTION

RM1,200

Confirm tier: RM100 a month, for 12 months

=

4 · OWNER GAINS

RM8,450

7.0 times what they paid

WIN BACK MORE OR LESS

Won back	RM a year	Return (ROI)	With admin time (ROV)
10%	3,860	2.2×	3.6×
25%	9,650	7.0×	8.5×
50%	19,300	15.1×	16.5×

Return = (money won back − RM1,200) ÷ RM1,200. With admin time also counts 20 hours saved at RM25.

3%

is all it takes. Winning back 3% of extras pays for the whole subscription.

One simulated job, so the pilot measures the real rate. Hours and hourly rate are assumptions: appendix A.

FUTURE

Built, tested, and what we validate next

PROVEN

- ✓ The full journey runs end to end
- ✓ 290+ automated tests pass
- ✓ Roles are enforced on the server
- ✓ A renovation chat runs unchanged
- ✓ AI reader beat our plain rules on blind chats: 17 vs 25 of 29

VALIDATION PLAN

- ✓ Blind chats from teammates: scored
- ✓ AI reader: re-tested on teammates' chats
- ✓ Live WhatsApp group: pilot
- ✓ RM100 price: tested with owners
- ✓ Real unbilled rate: measured in pilot

RISK AND SAFEGUARD

The AI reader still gets some chats wrong: 25 of 29 fully right

SAFEGUARD

A person checks every change and the client confirms before anything is billed.

Next 2 weeks: re-test the reader on chats written by teammates, time the bill by hand against SMErgers, and test the price with real business owners.

Test log and results: docs/results.md in the repository.

Roadmap

PHASE 1 · OCT 2026

Prove it

- Webhook and chat import
- Approvals, quote versions
- Suggestions from past jobs

PHASE 2 · Q1 2027

Pilot

- 3 to 5 event companies
- Official WhatsApp API
- Orbyte approvals and eSign

PHASE 3 · 2027 ONWARD

Package it

- Model trained on approvals
- Renovation, agencies, print
- Compare vendor quotes
- Client photo to change

Impact KPI: agreed changes that reach the bill, from an estimated 75% to 95% · final-bill time cut by half

SDG 8.2 · 8.3: small firms keep the revenue they have earned. Proof to earn: real chats, hours saved, disputes settled

THANK YOU

Catch it. Confirm it. *Sync it.*

TEAM VBALL

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SE Steven Elnathan Meliala

LW Looi Wan Wei

Orbyte Smart SME Challenge · Domain 2 · SDG 8.2 · 8.3

Scan for the code

github.com/Anonys-sky/SMErgers · Apache 2.0 · Live demo: sm-ergers-1.replit.app · Demo video: youtu.be/UKgKFRvKVhM

APPENDIX A

Every number: source, confidence, caveat

Number	Source	Confidence	Caveat
102,657 construction and renovation firms (market input)	DOSM Economic Census 2023	High	Whole sector, not only renovation
About 25,000 other project firms	Our assumption	Low	No public event-company count
20% run jobs over WhatsApp	Our assumption	Low	To measure in the pilot
35 messages, 10 changes, RM3,860 (11.2%) on one job	Synthetic dinner chat (data/chats)	Synthetic	One job; real leakage unknown
20 admin hours at RM25	Our assumption	Low	Owners to confirm
Competitor features (grid)	Vendor pages, search results	Medium	Front pages only; not tested
Changes at “almost every event”	Team member’s experience	Medium	One person, one company
Blind chats: AI 17 of 29, rules 13	docs/results.md, Runs 2, 5 and 6 (RunPod)	Low	Synthetic chats by AI agents; 29 changes. 21 after fixes, same chats (Run 6)

A widely repeated “52% of projects see scope creep” figure could not be traced to its source, so it is not used. Checked 21 to 24 Oct 2026.
dosm.gov.my · rentman.io · helloendless.com · workflowmax.com

APPENDIX B

Boundaries, privacy and tools

INSIDE THE BRIEF

- No payments or filings
- No legal or financial advice
- Bill is a draft, not an invoice
- Synthetic data only
- Sign-in simulated; roles still enforced

PDPA, IN REAL USE

- Tell people why the group is read
- Keep only messages tied to changes
- Limit access by role
- Set retention; honour requests
- Legal review before any pilot

TOOLS DECLARED

- Replit: build and host
- GitHub: public, Apache 2.0
- Claude, Replit Agent: AI coding help
- Llama 3.1 8B: the AI reader
- WhatsApp Cloud API: design
- RunPod: GPU that runs the AI reader (no training)

WhatsApp, honestly: Meta’s Groups API allows 8 people per group and needs a verified business. An unofficial client (whatsapp-web.js) was used only for team testing, never in the product.

Boundaries from the Domain 2 brief, section 6. Not legal advice.

APPENDIX C

Hard questions, plain answers

“Isn’t it just an invoice maker?”

No. It finds and confirms what was agreed. Invoicing stays in accounting software.

“How accurate is it?”

On three blind synthetic chats, 17 of 29 changes fully right (21 after we fixed the errors seen, same chats). Small sample, so teammates’ chats and a pilot come next.

“Is reading WhatsApp allowed?”

Through Meta’s official API, in groups the business creates, with consent.

“Will owners pay RM100?”

The pilot tests it. Break-even is 3% of extras recovered.

“Why not just write it down?”

Deals happen mid-chat at 11 pm. We read the chat they already use.

“Is the AI reader proven?”

Not on real chats yet. The pilot tests that.

“What if the client never confirms?”

It stays off the bill, on the director’s list. Nothing is billed silently.

“What if a message is misread?”

Unclear items go to a person. Nothing reaches the bill without a check and the client’s yes.

Rehearse these with a teammate playing a hostile judge.